

Unit – 1

Introduction

Data

It is the collection of raw facts and figures. E.g. length = 2, breadth = 3

Information

Processed data is known as information.

Data $\xrightarrow{\text{Processed}}$ Information

E.g. Area = length $\times$ breadth

Difference between Data and Information

Data	Information
It is raw or known facts.	It is processed or refined data.
It stores the facts.	It represents the facts.
It is technology based.	It is business based.
Data is gathered from various sources.	Information is transformed from data.
Data do not have fixed format.	Information is basically in the form of table graph, curve line, chart etc.

Quality of Information

- 1) **Accuracy:** Information must be free from errors. It should be clear and should reflect the meaning of data on which it is based.

Accuracy means it should be free from bias manipulated or distorted information is worse than no information.

- 2) **Timelines:** Information must be delivered to right time, right place and right person of the information is out-dated and not updated then those information are not important for future use.

- 3) **Relevance:** Information is good only if it is relevant, i.e. meaningful to the decision maker. Information relevant for one person may not be relevant for second person.

- 4) **Completeness:** Information should be complete. By complete we mean it should be updated. Information is proper decision to decision maker.

Database

- Database is location where we store data for future use.
- A database is a collection of related data stored in a medium.
- A **database** is an organized collection of structured information or data, typically stored electronically in a computer system.

The collection of data usually refer to database that contains information about a particular enterprise, organization school,

college etc. By data we mean known facts that can be recorded and that have implicit. For example: consider the name, telephone numbers and address of people that you know and for future use you can store them in index book, hard disk, pen drive etc.

Database+ management software= database management software

Database Management System (DBMS)

- A DBMS is a collection of inter-related data and a set of programs to access the data.
- A **Database Management System (DBMS)** is the specialized software that interacts with users, applications, and the database itself to capture and analyze the data.
- A DBMS is process of managing data in database. It allows user to create and maintain database. The primary goal of DBMS is to provide the way to store and retrieve the data from database.
- A database is a software a package or a system to facilitate the creation and maintain of a computerized database.

Example

Oracle, Sybase, Mysql, Sql Server, MS-Access etc.

- i) and effective.
- ii) Provide better access to more and better managed data.
- iii) Reduce the probability of inconsistent data.
- iv) Query language allows quick answer to ad-Hoc query.

- v) It helps to remove data redundancy.

Application of DBMS

- i) **Banking:** For maintaining customer information, accounts, loans and banking transactions.
- ii) **Universities:** For maintaining student record, course registration and grades.
- iii) **Railway Reservations:** For checking the availability of reservation in different trains, tickets, etc.
- iv) **Airlines:** For reservation ad schedule-information.
- v) **Tele-Communication:** Keeping records of all telephone calls, generating monthly bills etc.
- vi) **Finance:** For storing information about holidays, sales and purchase of financial instruments.
- vii) **Sales:** For customers, products and purchase information.
- viii) **Manufacturing:** For tracking production, inventory, orders, supply chain management.
- ix) **Human Resource:** For starting the information about employer records, salaries, tax and deduction.

Advantages of Database Management System

The advantages of database management systems are:

1. **Data Security:** DBMS enhances data security through access control and encryption. It enforces privacy policies and prevents unauthorized access. As user numbers grow, it mitigates associated risks of breaches.
2. **Data integration:** DBMS unifies data from different sources into a centralized system. This provides a coherent

organizational view of operations. It helps to keep track of how one segment of the company affects another segment.

3. **Data abstraction:** Since many complex algorithms are used by the developers to increase the efficiency of databases that are being hidden by the users through various data abstraction levels to allow users to easily interact with the system.
4. **Reduction in data Redundancy:** DBMS avoids data duplication by enforcing unique constraints. It removes unnecessary repetitive entries in databases. This ensures efficient use of storage and improves consistency. for e.g. - If there are two same students in different rows, then one of the duplicate data will be deleted.
5. **Data sharing:** A DBMS provides a platform for sharing data across multiple applications and users, which can increase productivity and collaboration.
6. **Data consistency and accuracy:** DBMS enforces integrity constraints to maintain valid data. It minimizes discrepancies by syncing updates across all views. This reduces errors and ensures reliable & consistent data.
7. **Data organization:** A DBMS provides a systematic approach to organizing data in a structured way, which makes it easier to retrieve and manage data efficiently.
8. **Efficient data access and retrieval:** DBMS allows for efficient data access and retrieval by providing indexing and query optimization techniques which reduces time taken to

retrieve large datasets. It boosts system performance and user satisfaction.

9. **Concurrency and maintained Atomicity:** That means, if some operation is performed on one particular table of the database, then the change must be reflected for the entire database. The DBMS allows concurrent access to multiple users by using the synchronization technique.
10. **Scalability and flexibility:** DBMS is highly scalable and can easily accommodate changes in data volumes and user requirements. It allows flexible schema modifications and expansion. This makes it ideal for dynamic organizational environments and can scale up or down depending on the needs of the organization.

Disadvantage of Database

1) Problem Associated with Centralization

Centralization increase the security problem and disturbance due to the down times and failures.

2) Cost of Software

Today's there are several software which are very costly. Hence, from economic point of view it is the drawback i.e. for very small organization it is not suitable.

3) Technically Qualified Human Resource

To manage and provide security to database technical knowledge should be required and employed should also have knowledge to use computer. So, it is expensive to have technical person such as database administrator.

4) Complexity of backup and Recovery

DBMS provides the centralization of data, which requires the adequate backup of the data so that in case of failure the data can be recover.

5) Cost of Hardware

For DBMS hardware should be latest and branded as DBMS is centralized networking cost increases.

The main characteristics of the database approach versus the file-processing approach are the following:

Characteristics of the Database Approach

1. Self-Describing Nature of a Database System

- A database system stores not only the data but also the metadata (data about data).
- Metadata includes structure, constraints, and relationships of the data.
- This makes the database self-describing, unlike traditional file systems where structure is hidden inside application programs.

2. Insulation Between Programs and Data (Data Abstraction)

- Programs are independent of how data is stored.
- Changes in storage structure or access methods do not require changes in application programs.
- Data abstraction: Users see logical view (what data), not physical view (how stored)

3. Support of Multiple Views of the Data

- Different users can see different perspectives of the same database.

Example:

- A student sees grades and courses.
- An administrator sees fees and enrollment details.
- Views provide security and simplicity by restricting access to only relevant data.

4. Sharing of Data and Multiuser Transaction Processing

- Databases allow concurrent access by multiple users.

Transaction processing ensures:

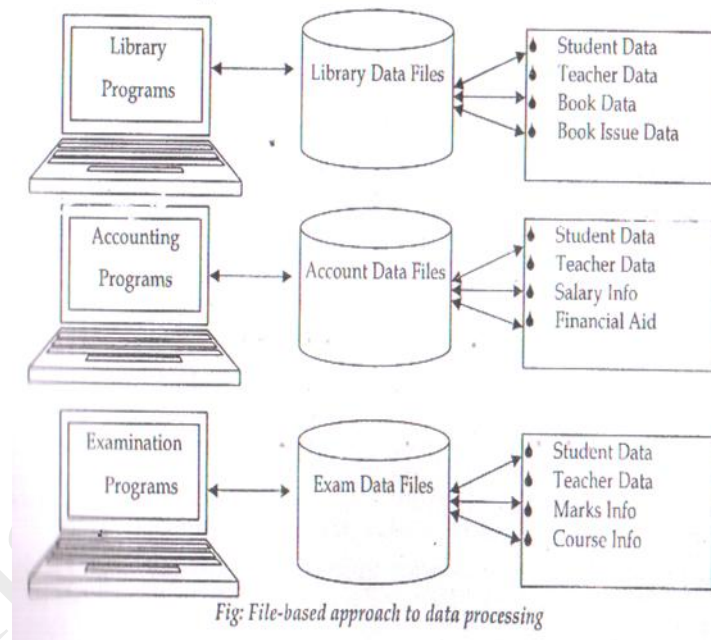
- Atomicity: All or nothing execution.
- Consistency: Database remains valid after transactions.
- Isolation: Transactions do not interfere with each other.
- Durability: Results are permanent after completion.

CREATE TABLE STUDENT

```
(
Rollnoint(3) primary key,
Name char(30) NOT null;
City char(20) default "Janakpur"
Mark float(10,2) NOT null
);
```

Database Management System vs. Flat File Systems

- File management system (FMS) is also called flat file systems. It stores data in a plain text file.
- A flat file is a file that contains records, and in which each record is specified in a single line. Fields from each record may simply have a fixed width with padding, or may be delimited by whitespace, tabs, commas or other characters.
- There are no structural relationships and the data are "flat" as in a sheet of paper.
- In this approach each application has data files related to it containing all the data records needed by the application. Thus, an organization has to develop number of application programs each with an associated application-specific data files.
- For example, in a computer system, the checking account processing system would have its own data files. This file based approach to data processing is shown in fig below:



Problems with Traditional File-Based Systems

Before the introduction of modern DBMS, data was managed using basic file systems on hard drives. While this approach allowed users to store, retrieve and update files as needed, it came with numerous challenges

- **Data Redundancy:** Duplicate entries across files
- **Inconsistency:** Conflicting or outdated information
- **Difficult Access:** Manual file search required
- **Poor Security:** No control over data access
- **Single-User Access:** No support for collaboration

- **No Backup/Recovery:** Data loss was often permanent

Advantages of Database Management System over Traditional File System

1. **Better Data Security:** DBMS offers centralized control with advanced security features like encryption, authentication, and authorization. It restricts unauthorized access and protects data from internal and external threats. This ensures sensitive information is kept secure and confidential. Compared to file systems, DBMS provides a much stronger security model.
2. **Reduced Data Redundancy:** DBMS stores data in a centralized and structured way, minimizing duplication. It supports shared access across applications, eliminating repeated storage. Less redundancy means less storage use and fewer inconsistencies. File systems lack such control and often store duplicate data across files.
3. **Improved Data Consistency:** Data consistency is maintained by enforcing validation rules and constraints. Changes in data reflect across all applications accessing it. This ensures uniformity of data for every user and use case. Traditional systems lack synchronization, causing discrepancies.
4. **Improved Data Access and Availability:** DBMS provides efficient data access and retrieval mechanisms that enable quick and easy data access. It allows multiple users to access the data simultaneously, ensuring data availability.
5. **Improved Data Sharing:** DBMS allows secure and simultaneous data sharing across users and systems. It supports real-time collaboration without conflict or data loss. Departments can interact seamlessly using a common data source. Traditional systems often require data duplication for sharing.
6. **Improved Data Integration:** DBMS combines data from various sources into a unified view. It maintains integrity through referential, domain, and entity constraints. This enables accurate analysis and decision-making organization-wide. File systems struggle to integrate data from disparate platforms.
7. **Improved Data Backup and Recovery:** DBMS offers automated, reliable backup and point-in-time recovery. Data can be restored after failures, maintaining consistency and availability. It supports disaster recovery strategies, ensuring business continuity. Traditional file systems often require manual, error-prone backups.

8. **Data independence:** DBMS separates data's logical structure from its physical storage. Users can access and manipulate data without knowing its location. This abstraction improves flexibility and reduces dependency on hardware/software changes. File systems tightly bind data to physical structure, limiting adaptability.

DBMS	File processing system
1. A Database Management System (DBMS) is a collection of interrelated data and the set of programs to access those data. 2. Data redundancy problem is not found. 3. Data inconsistency does not exist. 4. Accessing data from database is easier. 5. The problem of data isolations is not found. 6. Atomicity and integrating problems are not found. 7. Data are more secure. 8. Concurrent access and crash recovery.	1. A flat file system stores data in a plain text file. A flat file is a file that contains records, and in which each record is specified in a single line. 2. Data redundancy problem exist. 3. Data inconsistency may exist. 4. Accessing data from database is comparatively difficult. 5. Here data are scattered in various files and formats so data isolation problem exist. 6. Here these problems are found. 7. Data are less secure. 8. Here there is no concurrent access and no recovery.

ACTORS ON THE SCENE

Actors on the scene are individuals who directly interact with and manage databases in daily operations. These roles ensure efficient database use, design, and application development.

The people whose jobs involve the day-to-day use of a large database are called actors on the scene.

Database Administrator

a. Database Administrator (DBA)

- The DBA is the person or team responsible for implementing and managing the database system in an organization.
- The DBA has all the privileges allowed by the DBMS.
- The DBA can grant access rights to users and revoke them when needed.
- The DBA controls security, ensuring only authorized users can access or modify data.
- The DBA also handles backup, recovery, maintenance, and performance monitoring of the database.
- The DBA is also accountable for problems such as security breaches and poor system response time.

b. Database Designers

- Database designers are responsible for identifying the data to be stored in the database and for choosing appropriate structures to represent and store this data.
- It is the responsibility of database designers to communicate with all prospective database users in order to understand their requirement and to create a design that meets the requirements.
- The database design must be capable of supporting the requirements of all user groups.

c. End Users:

End users are the people whose jobs require access to the database for querying, updating, and generating reports. There are several categories of end users:

1. Causal end users : They occasionally access the database, but they may need different information each time. They use a sophisticated database query interface to specify their requests.
2. Naive end users : They constantly query and update the database using standard types of queries and updates called canned transactions that have been carefully programmed and tested. (e.g., ATM systems).
3. Sophisticated end users : It includes engineers, scientists, business analysts and others in order to meet their complex requirements.
4. Standalone users : They maintain personal databases by using ready-made, program packages that provide easy-to-use menu-based or graphics-based interfaces.

- Tool developers include persons who design and implement tools-the software packages that facilitate database system design and use and that help improve performance.
- Operators and maintenance personnel are the system administration personnel who are responsible for the actual running and maintenance of the hardware and software environment for the database system.

WORKERS BEHIND THE SCENE

- DBMS system designers and implementers are persons who design and implement the DBMS modules and interfaces as a software package.